

Shakti Group Of Companies

Integrated manufacturer and supplier of industrial & specialty gases

www.shaktigases.com

Our Group Of Companies

Shaktishree Air Private Limited (SAPL)	Manufacturing plant of Liquid Oxygen / Nitrogen / Argon Gas Kandari, Karjan Vadodara Gujarat) 178 TPD Capacity Per
Shakti Industrial Gases	Supplier of Argon, Oxygen, Nitrogen and Carbon-Dioxide. Plant located at: Plot No. 628/14, G.I.D.C, Manjusar, Ta. Savli, Dist: Vadodara- 391 350
Shakti Air Products	Leading Supplier of Argon, Nitrogen, Carbon-dioxide & Oxygen. Plant located at: Plot No. 2 Shri Hari Industrial Park, Por GIDC, Por - Ramangamdi Road Por, Vadodara- 391 243
Shakti Gases	Leading Supplier of Hydrogen Gas in Cylinder(s), Cylinder Skids and Cascades. Plant Located at: 1002, GIDC Industrial Estate, RANOLI, Dist: Vadodara- 391 350
Shreeji Acetylene Co.	Leading Manufacturer of Dissolved Acetylene Gas. Plant located at: Plot No. 37. G.I.D.C. Alindra, Ta:Savli, Dist: Vadodara.
Moxi Industrial Gases	Supplier of Carbon Dioxide Gas. Plant located at: Survey No.: 431/1/P, Village: Moxi, Ta: Savli, Dist:Vadodara.
Shreeji Acetylene Co (2) .	Leading Manufacturer of Refilling Station of Oxygen, Nitrogen, Argon, and Co2 Gas Cylinders. Plant located at: Palsana, Gujarat.
Indore Carbonic	Leading Manufacturer of Plant of Liquid carbon dioxide and our plants are located at Barwaha, Indore (M.P) / Lebad, Indore (M.P) / Ahmedabad / Hyderabad
Bhavani Enterprise	Leading Dry Ice , Ethylne Oxide Distributor all over India and Exports • 45, Maruti Complex, Ranoli GIDC, Ranoli, Dist: Vadodara-391350, Gujarat • Our Dry ICE Outlets - Vadodara / Ahmedabad / Surat / Rajkot





Our Core Products

Oxygen, Argon, Nitrogen, Carbon Dioxide & Hydroden



Bulk Tankers gases supply

Our Bulk Tanker Gas Supply solutions provide safe, efficient, and reliable delivery of industrial and specialty gases in large volumes to meet the demanding needs of various industries. With a modern fleet of high-capacity cryogenic and compressed gas tankers, we ensure uninterrupted supply and strict compliance with international safety and quality standards.



Refilling Cylinder Gases

Our Cylinder Gas Refilling Services provide safe, accurate, and reliable refilling of industrial, medical, and specialty gas cylinders. With advanced filling systems and strict quality control procedures, we ensure every cylinder meets the highest safety and purity standards before delivery.



Gases Tank Installation

Our Gas Tank Installation Services provide complete, safe, and efficient setup of bulk gas storage systems for industrial, medical, and commercial applications. From site assessment to commissioning, we deliver turnkey solutions tailored to your operational requirements while ensuring full compliance with safety and regulatory standards.



Purging JOB

Our Gas Purging Services ensure the safe removal of unwanted or hazardous gases from pipelines, storage tanks, vessels, and industrial systems before maintenance, commissioning, or shutdown activities. We use controlled purging methods to eliminate oxygen, moisture, and contaminants, creating a safe and inert environment for operations.



Bulk Tankers gases supply

Our Capablities

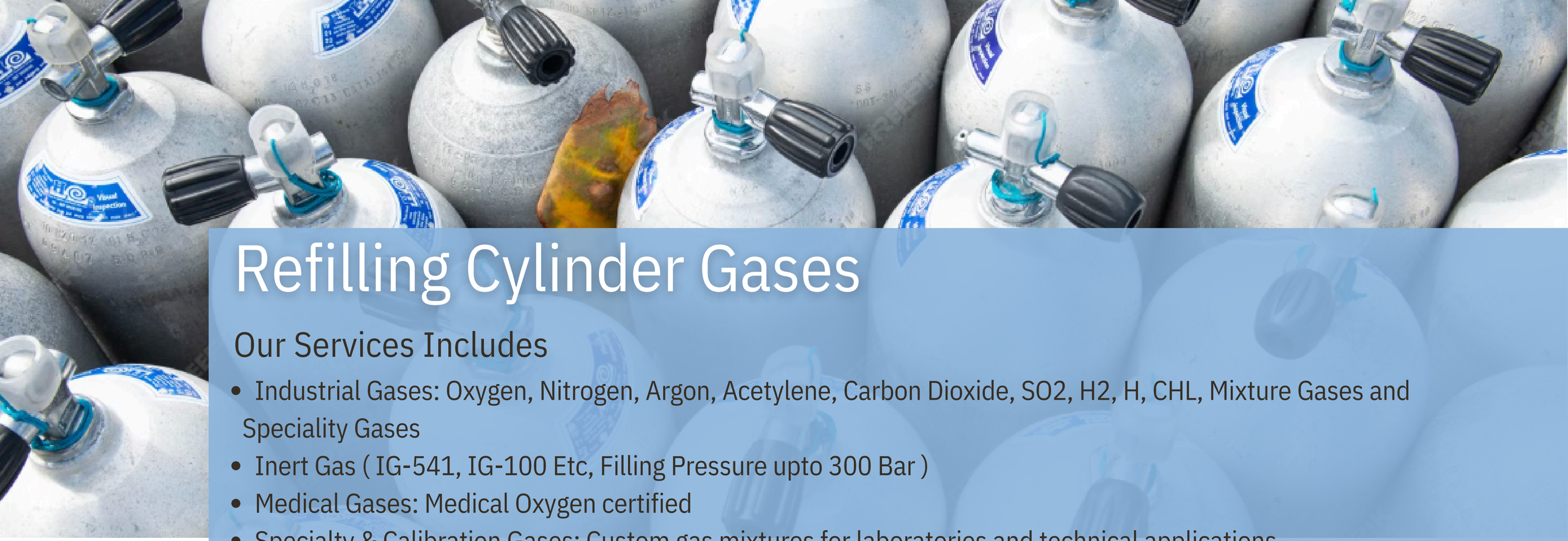
- Cryogenic Liquid Gases: Oxygen (LOX), Nitrogen (LIN), Argon (LAR), Carbon Dioxide (LCO_2)
- Compressed Gases: Hydrogen, Helium, Specialty and Industrial Gas Blends
- High-Capacity Deliveries: Optimized logistics for bulk consumption facilities
- On-Site Storage Solutions: Installation and maintenance of bulk tanks and vaporization systems
- 24/7 Monitoring & Support: Real-time supply management and emergency response

Industries we Serve in

- Healthcare & Medical Facilities
- Oil & Gas
- Petrochemical & Refineries
- Manufacturing & Fabrication
- Food & Beverage Processing
- Power Generation
- Steel & Metallurgy

Why Choose Us?

- Advanced tanker fleet with certified safety systems
- Reliable and timely deliveries
- Competitive pricing and long-term supply contracts
- Technical support and customized gas management solutions
- Commitment to safety, quality, and environmental responsibility
- Our bulk gas supply services are designed to optimize operational efficiency, reduce downtime, and ensure consistent product quality for your business. Partner with us for dependable gas solutions that power your operations forward.



Refilling Cylinder Gases

Our Services Includes

- Industrial Gases: Oxygen, Nitrogen, Argon, Acetylene, Carbon Dioxide, SO2, H2, H, CHL, Mixture Gases and Speciality Gases
- Inert Gas (IG-541, IG-100 Etc, Filling Pressure upto 300 Bar)
- Medical Gases: Medical Oxygen certified
- Specialty & Calibration Gases: Custom gas mixtures for laboratories and technical applications
- Cylinder Inspection & Testing: Hydrostatic testing, valve inspection, and safety compliance checks
- Pickup & Delivery Services

Safety & Quality Assurance

- Precision filling with calibrated systems
- Compliance with international safety standards
- Leak testing and cylinder integrity checks
- Proper labeling and traceability for every cylinder

Industries We Serve

- Hospitals & Healthcare
- Manufacturing & Fabrication
- Oil & Gas
- Laboratories & Research Centers
- Food & Beverage
- Construction & Engineering



Gases Tank Installation

Our Installation Solutions Includes

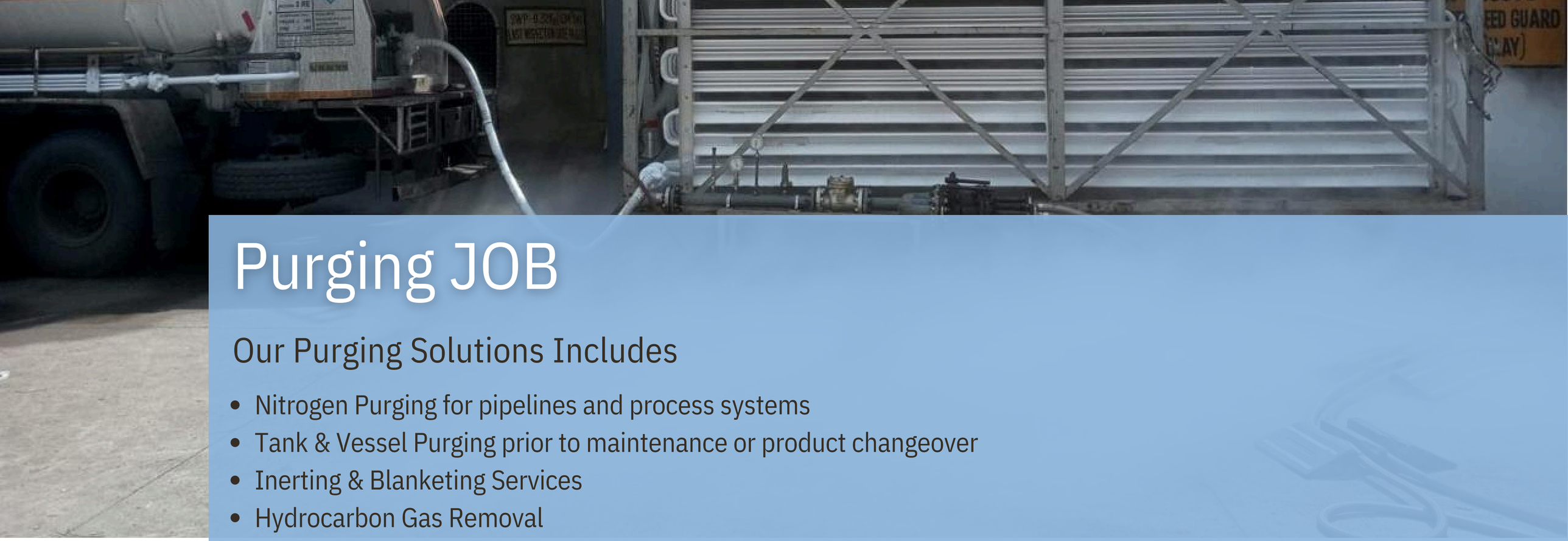
- Cryogenic Storage Tanks: Liquid Oxygen (LOX), Liquid Nitrogen (LIN), Liquid Argon (LAR)
- Pressure Vessels & Storage Tanks
- Vaporizer Systems Installation
- Pipeline & Gas Distribution Network Setup
- Control Panels & Monitoring Systems
- Safety Systems

Our Process

- Precision filling with calibrated systems
- Compliance with international safety standards
- Leak testing and cylinder integrity checks
- Proper labeling and traceability for every cylinder

Industries We Serve

- Hospitals & Healthcare
- Manufacturing & Fabrication
- Oil & Gas
- Laboratories & Research Centers
- Food & Beverage
- Construction & Engineering



Purging JOB

Our Purging Solutions Includes

- Nitrogen Purging for pipelines and process systems
- Tank & Vessel Purging prior to maintenance or product changeover
- Inerting & Blanketing Services
- Hydrocarbon Gas Removal
- Pre-Commissioning & Decommissioning Purging
- Pressure Testing & Leak Testing Support

Applications

- Oil & Gas Facilities
- Refineries & Petrochemical Plants
- Power Generation Plants
- Manufacturing & Processing Units

Safety & Compliance

- Experienced technicians and safety supervisors
- Controlled pressure and flow management
- Compliance with ISO safety standards
- Detailed job reporting and documentation

Shaktishree Air Private Limited



Dedicated Medical delivery system and Healthcare Team

- **Medical Oxygen Supply** :

High-purity Liquid Oxygen as per IP 2010 supports hospitals and healthcare facilities, providing life-saving respiratory support for patients.

VACUUM INSULATED SUPPLY TRANSPORT TANKERS FOR LIQUID NITROGEN, LIQUID OXYGEN, LIQUID ARGON, LIQUID CO2







Bulk Gases supply & Applications



Acetylene C_2H_2

Ethyne

CAS: 74-86-2
EC: 200-816-9
UN: 1001

AUTO CHEM CONSTR FOOD MANUF MEDICAL OEM PETRO R&D SEMI

Purity grade	Typical purity	Typical impurities [ppm]		
		Air	PH□	H□S
HiQ® Acetylene 2.6 AAS	>99.6 %	≤4,000	<5	<1

Typical filling pressure: 15 °C: 15 bar(g)/ 70 °F: 250 psi(g)

Typical packages

Cylinders	Bundles	Drum tanks	ISO tanks	Tube trailer	Road tanker
•	•				

Typical ancillary equipment

Pressure control valves	Gas distribution panels/manifolds	Liquid flow control valves	Customised distribution systems
•	•		Consult local team

Characteristics
Flammable. Colourless gas with ether-like odour when very pure, otherwise garlic-like. Supplied dissolved in acetone or DMF (n,n-dimethylmethanamide). Can decompose instantaneously at pressures higher than 1 bar. Acetylene can be delivered as a non-dissolved gas for specific R&D applications. Gas density is slightly lighter than air.

Source
Acetylene is manufactured commercially by reaction between calcium carbide and water, and as a by-product of ethylene production.

Applications
Acetylene is used as a raw material for the production of electrically conducting plastics, such as polyacetylene.

Acetylene is used with high purity synthetic air or nitrous oxide as a fuel for the flame in atomic absorption flame spectroscopy. This is used in water, soil, food and biological research laboratories where sensitivity and accuracy of results are important.

Acetylene is most commonly used in combination with oxygen for cutting or welding materials such as mild steel, where the standard industrial grade is sufficient. Acetylene with low phosphine levels is required for lead brazing or welding.

Acetylene is used in organic synthesis (laboratory work) as well as in chemical synthesis.

Acetylene is used as carbon source in the production of molecules like fullerenes; well-known examples are buckyballs or carbon nanotubes.

Acetylene is used in the cultivation of plants; it improves the formation of new flowers.

Acetylene is used as a component in calibration gases for the gas, oil and chemical industries.

Acetylene is still used in some lighthouses as light fuel source.

Acetylene is one of the components of lung testing gases.

This unsaturated hydrocarbon exhibits high chemical reactivity, and is an important intermediate in the chemical industry. It is employed for the production of:
→ acetaldehyde
→ acrylic acids
→ acrylic ethers
→ acrylonitride
→ carbazole
→ butenyne (vinyl acetylene)
→ chloroethene (vinyl chloride)
→ diols
→ ethene
→ ethenoxyethenes (vinyl ethers)
→ ethenyl acetate (vinyl acetate)
→ ethenyl amides (vinyl amides)
→ ethenyl sulfides (vinyl sulfides)
→ neoprene
→ phenylethene (styrene)
→ polyoxymethylene
→ pyrrolidine
→ trichloroethene
→ very fine carbon black, called “acetylene black”.

Hazard classifications
Globally Harmonized System of classification of chemicals (GHS)
GHS-CLP
Signal word: DANGER



H-statements:
Dissolved Gas → H280 – Contains gas under pressure; may explode if heated; H220 – Extremely flammable gas; H230 – May react explosively even in the absence of air.

Transport of dangerous goods



ADR Class 2, 4F



DOT Class 2.1

Oxygen O_2

CAS: 7 782-4 4-7
EC: 231-956-9
UN: 1072 (Compressed) 1073 (Refrigerated liquid)
UN: R-732

AUTO CHEM CONSTR ENERGY FOOD MANUF MEDICAL METAL OEM PETRO PHARMA R&D SEMI

Purity grade	Purity	Impurities [ppm]						
		H_2O	CnHm	CO	CO_2	N_2	Ar	Odour
HiQ® Oxygen 3.5	≥99.95 %	≤5	-	-	-	-	-	-
HiQ® Oxygen 4.8	≥99.998 %	≤3	≤1	-	-	≤10	≤10	-
HiQ® Oxygen 5.0 Zero	≥99.999 %	≤3	≤0.2	≤1	≤1	≤5	≤5	-
HiQ® Oxygen 6.0	≥99.9999 %	≤0.5	≤0.1	≤0.1	≤0.1	≤0.5	≤1	-
VERISEQ® Process Oxygen (pharmaceutical grade)	≥99.5 %	≤67	-	≤5	≤300	-	-	N/D

Typical filling pressure: 15 °C: 200 bar(g)/ 70 °F: 2,800 psi(g)

Typical packages

Cylinders	Bundles	Drum tanks	ISO tanks	Tube trailer	Road tanker
•	•				Cryogenic liquid

Typical ancillary equipment

Pressure control valves	Gas distribution panels/manifolds	Liquid flow control valves	Customised distribution systems
•	•	Cryogenic liquid	Consult local team

Characteristics
Colourless and odourless gas. Many materials burn in oxygen that do not normally burn in air. Reduces the flash-point temperature and increases the combustion speed. Gas density is heavier than air.

Hazard classifications
Globally Harmonized System of classification of chemicals (GHS)
GHS-CLP
Signal word: DANGER



H-statements:
Compressed Gas → H280 – Contains gas under pressure; may explode if heated; Refrigerated Gas → H281 – Contains refrigerated gas; may cause cryogenic burns or injury; H270 – May cause or intensify fire; oxidiser.

Transport of dangerous goods



ADR Class 2, 1O (Compressed)
3O (Refrigerated liquid)



DOT Class 2.2

Source Oxygen is obtained on a commercial scale by the liquefaction and subsequent distillation of air. For very high purity oxygen it is normally necessary to take the product from an air separation plant through a secondary purification and distillation stage. Alternatively high-purity oxygen may be produced by the electrolysis of water. Lower purities of oxygen can also be produced with membrane technique.

Applications
Many oxidation reactions in the chemical industry use pure oxygen rather than air in order to benefit from higher reaction rates, easier product separation, higher yields, or smaller equipment size.
High-purity oxygen is used for the formation of silicon dioxide and metal oxide, as an etchant for photoresist, and in mixtures with halocarbons for etching silicon. Oxygen is also used in conjunction with hydrogen to fuel torches for welding, brazing, glass blowing and tube sealing for a variety of electronic components such as reed relay switches.

High-purity oxygen is used in conjunction with high-purity methane in Advanced Gas Cooled (AGR) nuclear reactors to maintain an appropriate carbon balance in the (CO_2) gas coolant in the nuclear core.

High-purity oxygen is used in the optical fibre production process.

Injecting oxygen into sewage treatment plants accelerates the decomposition of sewage.

Oxygen is used for chemical synthesis.

Oxygen is used as an oxidiser.

Oxygen is used to supplement or replace air in burners used in many different industries in order to obtain increased temperatures. Typical applications are found in the steel, non-ferrous, glass and concrete industries amongst many others.

Oxygen is used for flame sealing of glass ampoules for finished products for the pharmaceutical industry and the chemical industry.

In the food industry, oxygen is used in the transportation of live fish and seafoods.

Oxygen is used for enrichment of air during fermentation.

Mixed with other gases, oxygen serves in the production of breathable atmospheres (O_2 + CO_2: reanimation; O_2 + He or O_2 + N_2: underwater diving).
Oxygen is used in some cases for modified atmosphere packaging (MAP) of food stuffs. It is used either pure or in mixtures with carbon dioxide and/or nitrogen.
Liquid oxygen is used in liquid oxygen explosives, and as a comburent in space propulsion.

Oxygen is used in the medical field, as pure gas and in mixtures.

Oxygen is also used in calibration gas.

Oxygen is used in metal treating laser applications. Oxygen is used in cutting and welding.

Argon Ar

CAS: 7440-37-1
EC: 231-147-0
UN: 1006 (Compressed)
UN: 1951 (Refrigerated liquid)
R -74 0

AUTO CONSTR FOOD MANUF MEDICAL METAL OEM PHARMA R&D SEMI

Purity grade	Purity	Impurities [ppm]								
		H_2O	O_2	CnHm	CO	CO_2	N_2	H_2	CH_4	Halocarbons
HiQ® Argon 4.8	≥99.998 %	≤5	≤5	-	-	-	≤10	-	-	-
HiQ® Argon 5.0	≥99.999 %	≤3	≤2	≤0.5	-	-	≤5	-	-	-
HiQ® Argon 5.0 Zero	≥99.999 %	≤3	≤2	≤0.2	≤1	≤1	≤5	-	-	-
HiQ® Argon 6.0	≥99.9999 %	≤0.5	≤0.5	≤0.1	≤0.1	≤0.1	≤0.5	≤0.5	-	-
HiQ® Argon 7.0 VERISEQ® Process Argon	≥99.99999 % ≥99.998 %	≤50 ppb	≤30 ppb	≤30 ppb	≤30 ppb	≤30 ppb	-	≤30 ppb	-	≤1 ppb
(pharmaceutical grade)		≤0.5	≤5	-	-	-	≤10	-	≤1	-

Typical filling pressure: 15 °C: 200 bar(g)/ 70 °F: 2,800 psi(g)

Typical packages

Cylinders	Bundles	Drum tanks	ISO tanks	Tube trailer	Road tanker
•	•				Cryogenic liquid

Typical ancillary equipment

Pressure control valves	Gas distribution panels/manifolds	Liquid flow control valves	Customised distribution systems
•	•	Cryogenic liquid	Consult local team

Characteristics
Colourless and odourless gas. Non-reactive. Inert. Asphyxiant in high concentrations. Gas density is heavier than air.

Hazard classifications
Globally Harmonized System of classification of chemicals (GHS)
Proposed by the Industry
Signal word: WARNING



H-statements:
Compressed Gas → H280 – Contains gas under pressure; may explode if heated; Refrigerated Gas → H281 – Contains refrigerated gas; may cause cryogenic burns or injury; EIGA-As – Asphyxiant in high concentrations.

Transport of dangerous goods



ADR Class 2, 1A (Compressed)
3A (Refrigerated liquid)



DOT Class 2.2

Source
The most common source of argon is an air separation plant. Air contains approx. 0.93% (vol.) argon. A crude argon stream containing up to 5% oxygen is removed from the main air separation column via a secondary (“side-arm”) column. The crude argon is then further purified to produce the various commercial grades required. Argon may also be recovered from the exhaust streams of certain ammonia plants.

Applications
Argon is one of the most common carrier gases in gas chromatography. Argon is used as a carrier gas in sputtering, plasma etching and ion implantations, and as a blanket atmosphere in crystal growth.

Argon is also the choice gas for ICP spectroscopy (Inductively Coupled Plasma spectroscopy).

One of the most common applications of argon, either pure or in various mixtures, is as a shielding gas for arc welding.

Many Geiger-counting tubes contain argon or argon mixed with organic vapours or other gases, for example 10% methane in argon.

Argon is one of the principal gases used for filling incandescent (filament) lamps, generally in a mixture with nitrogen, krypton or neon, for phosphorescent tubes in mixtures with neon, helium and mercury vapour and for thyratron radio tubes, in mixtures with neon.

The argon-oxygen decarburising (AOD) process is the most common method of refining stainless steel, and uses large quantities of both gases supplied either in liquid form or via pipeline from an on-site plant.

The pharmaceutical industry uses argon to displace oxygen in the top of intravenous drug containers, extending product shelf-life

Liquid argon is used in cryosurgery, e.g. cryoablation to destroy cancer cells.

Argon, R-740, is used in gas mixtures for non-CFC ultra-low temperature refrigeration applications.

Argon:
- → is used in atomic absorption spectrometry as a blanket gas in the graphite furnace.
- → is used in blends with, for example, fluorine and helium in excimer lasers.
- → is used as an insulation gas in high-efficiency multi-pane windows to improve thermal insulation.
- → is used in the iron and steel industry to prevent oxidation of molten metals and alloys and for degassing and desulphurization of molten steel and iron baths.
- → is used, often in a mixture with hydrogen, as a protective atmosphere for the heat treatment of certain metals, particularly those which are susceptible to nitriding when treated in a nitrogen-based atmosphere. This includes stainless steels and many different specialised and therefore small-scale applications.
- → is used for wine preservation to eliminate air by the heavier argon, to prevent oxidation and extend the product quality for opened bottles and barrels.
- → is, sometimes in combination with nitrogen, used to inflate airbags.
- → is used, often in combination with nitrogen and/or carbon dioxide, as a clean fire extinguishing gas, since the inert properties do not damage any materials extinguished.
- → is used in laboratory as purge gas or balance gas in gas mixtures.

Carbon dioxide CO_2

R-744

CAS: 124-38-9 EC: 204-696-9
UN: 1013 UN: 2187
(Refrigerated liquid) R -74 4

AUTO CHEM CONSTR ENERGY FOOD MANUF MEDICAL METAL OEM PETRO PHARMA R&D SEMI

Purity grade	Purity	Impurities [ppm]														
		H_2O	O_2	N_2	CO	NH_3	NOx	H_2S	SO_2	O_2+N_2	CH_4	THnm CHnm (as CH_4)		(as C_6H_{12}) Total Sulfur	Non-volatile residue	Halocarbons
HiQ® Carbon dioxide 4.0	≥99.99 %	≤10	≤10	≤50	-	-	-	-	-	-	-	-	-	-	-	-
HiQ® Carbon dioxide 4.5	≥99.995 %	≤5	≤5	≤10	-	-	-	-	-	-	≤2	-	-	-	-	-
HiQ® Carbon dioxide 4.5 SFC	≥99.995 % ≥99.999 %	≤1 ≤2	≤5 ≤2	- ≤5	- ≤1	- -	- -	- -	- -	- -	- ≤1	≤5	≤50 ppb	-	≤1	≤10 ppb
HiQ® Carbon dioxide 5.0	≥99.999 %	≤1	≤2	-	-	-	-	-	-	-	-	-	-	-	-	-
HiQ® Carbon dioxide 5.0 SFE	≥99.5 %	≤67	-	-	≤5	≤25	≤2	≤1	≤2	-	-	≤1	≤10 ppb	-	≤ -	12 ppb
VERISEQ® Process												-	-	≤1		-
Carbon dioxide (pharmaceutical grade) VERISEQ® Research	≥99.99 %	≤10	-	-	≤5	≤25	≤2	≤1	≤2	≤50	-	-	-	≤1	-	-
Carbon dioxide (pharmaceutical grade)																

Typical filling pressure: 15 °C: 56 bar(g)/70 °F: 839 psi(g)

Typical packages

Cylinders	Bundles	Drum tanks	ISO tanks	Tube trailer	Road tanker
•	•				•

Typical ancillary equipment

Pressure control valves	Gas distribution panels/manifolds	Liquid flow control valves	Customised distribution systems
Gaseous Withdrawal	Gaseous Withdrawal	Liquid Withdrawal	Consult local team

Characteristics
Liquefied, colourless gas. Asphyxiant in high concentrations. Gas density is heavier than air.

Hazard classifications
Globally Harmonized System of classification of chemicals (GHS)
Proposed by the Industry
Signal word: WARNING



H-statements:
Compressed Gas → H280 – Contains gas under pressure; may explode if heated; Refrigereted Gas → H281 – Contains refrigerated gas; may cause cryogenic burns or injury; EIGA-As – Asphyxiant in high concentrations.

Transport of dangerous goods



ADR Class 2, 2A
3A (Refrigerated liquid)



DOT Class 2.2

Source
Carbon dioxide is recovered from many different sources. It is obtained as an off-gas from fermentation processes, lime-stone kilns, natural CO_2 springs, as well as gas streams from chemical and petrochemical operations. Recently, CO_2 is also recaptured from the off-gas from power plants.

Applications
Carbon dioxide is used extensively as a neutralizing agent for pH control, for example, in cement curing water treatment and in many other commercially important chemical applications.

Carbon dioxide is used as media for supercritical fluid extraction (SFE) in sample preparation and as a carrier gas for analytical and preparative supercritical fluid chromatography (SFC).

Carbon dioxide is used in many consumer products ranging from aerosol packaging to air guns that require pressurised gas because it is inexpensive and non-flammable; in the operation of pneumatic equipment and for the transfer of hazardous and flammable liquids.

Liquid carbon dioxide is becoming increasingly used as a refrigerant in mechanical refrigerating systems due to its environmental credentials. It has the ASHRAE number R-744.
"Dry ice", or solid CO_2 is commonly used for refrigeration.
Liquid/solid carbon dioxide is used for cooling gas chromatography ovens.

Possible refrigerant for MAC (Mobile Air Conditioning) due to European phase out of tetrafluoroethane (R-134a).

Owing to its stimulating effect on the nerve centres, carbon dioxide is employed in medicine in mixtures with oxygen, for reanimating victims of asphyxiation. It also serves in the treatment of certain skin affections.

A substantial volume of carbon dioxide is used for carbonating beverages such as beer and many soft drinks and conservation of wine, unfermented grape juice and various fruit juices.

Carbon dioxide is used to modify atmospheres, for example in green houses where it increases plant growth rates or combined with nitrogen to prolong quality in food packaging applications (MAP – Modified Atmosphere Packaging).
Carbon dioxide, when mixed with helium and nitrogen, is used as the active medium in carbon dioxide lasers.

Carbon dioxide is used as an inerting agent for various mild steel welding operations, often in combination with argon.

Carbon dioxide:
→ is used for the chemical vapour deposition of silicon dioxide.
→ is used for foam blowing. → is used in Coleman nitrogen analysers. → is used in mixtures for car emission monitoring and environmental monitoring.
→ is used for fire extinguishing.
→ is often used in combination with ethylene oxide for sterilizing purposes.
→ is also used for blood analysis and dehydration of penicillin.
→ is used for production of paints and varnishes.

Note:
Carbon dioxide is listed in the Kyoto Protocol, an international Framework Convention with the objective of reducing greenhouse gases.

Hydrogen H_2

CAS: 1333-7α-0
EC: 21β-γ0β-7
UN: 10α9 (Compressed)
UN: 19γγ (Refrigerated liquid)
R-702

Purity grade	Purity	Impurities [ppm]						
		H_2O	O_2	C_nH_m	CO	CO_2	N_2	Halocarbons
HiQ® Hydrogen 4.6	≥99.99γ %	≤β	≤β	-	-	-	-	-
HiQ® Hydrogen 5.0	≥99.999 %	≤3	≤2	≤0.β	-	-	≤β	-
HiQ® Hydrogen 5.0 Zero	≥99.999β %	≤3	≤2	≤0.2	≤1	<1	≤β	-
HiQ® Hydrogen 5.5 ECD	≥99.9999 %	≤0.β	≤0.β	≤0.1	≤0.β	≤0.β	-	≤1
	≥99.99999 %			≤0.1	≤0.1	≤0.1	≤0.β	ppb
HiQ® Hydrogen 6.0							-	-
HiQ® Hydrogen 7.0		≤β0 ppb	≤30 ppb	≤30 ppb	≤30 ppb	≤30 ppb		-

Typical filling pressure: 1β °C: 200 bar(g)/ 70 °F: 2,800 psi(g)

Typical packages

Cylinders	Bundles	Drum tanks	ISO tanks	Tube trailer	Road tanker
•	•			•	

Typical ancillary equipment

Pressure control valves	Gas distribution panels/manifolds	Liquid flow control valves	Customised distribution systems
•	•		Consult local team

Characteristics
Flammable. Odourless and colourless gas. Gas density is lighter than air.

Hazard classifications
Globally Harmonized System of classification of chemicals (GHS)
GHS-CLP
Signal word: DANGER



H-statements:
Compressed Gas → H280 – Contains gas under pressure; may explode if heated; Refrigerated Gas → H281 – Contains refrigerated gas; may cause cryogenic burns or injury; H220 – Extremely flammable gas.

Transport of dangerous goods



ADR Class 2, 1F (Compressed)
3F (Refrigerated liquid)



DOT Class 2.1

CHEM ENERGY FOOD MANUF METAL OEM PETRO PHARMA R&D SEMI

Source
Hydrogen is most frequently produced for on-site usage by steam reforming of natural gas. Such plants may also be used as sources of hydrogen for the merchant market. Other sources are the chlor-alkali process that electrolyses sodium chloride solution to produce chlorine, and various waste gas recovery plants, such as at oil refineries or steel plants (coke oven gas). Hydrogen is also produced by electrolysis of water.

Applications
High-purity hydrogen finds widespread usage in the electronics industry as a reducing agent and as a carrier gas.

High-purity hydrogen is used as a carrier gas in gas chromatography.

Hydrogen finds some usage in the welding and cutting of metals.

Hydrogen is used in large quantities, (bulk supply or on- site generation) for the hydrogenation of vegetable and animal oils to produce margarine and other fats, hydro- treatment of petroleum products, and hydrosulfuration of fuels in order to eliminate sulfur.

Hydrogen in large quantities is used in petrochemical processes that include hydrodealkylation, hydrodesulfurisation, hydrotreatment.

Hydrogen is used in leak testing applications.
Hydrogen is used in HF/DF chemical lasers (see page 121).
Hydrogen is used extensively in the metals industries because of its ability to reduce metal oxides and prevent oxidation of metals during heat treatment. It may be used either pure, as is often the case when heat treating stainless steel, or in a mixture with inert gases, argon or nitrogen. It is used in the production of carbon steels, special metals and semiconductors.

Hydrogen is used for combustion;
→ in industry, it is used to supply oxygen-hydrogen torches for glass working (quartz, Pyrex®, etc), in the fabrication of artificial precious stones (ruby, etc), and for under water oxycutting
→ in the laboratory, it is used in analyzer flames, reducing flame photometry detection instruments, flame ionisation detection instruments, and fuel cells.

Extremely pure hydrogen is used in the chemical industry for fine reduction processes.

Liquefied hydrogen is used as a rocket fuel. In the laboratory liquid hydrogen is employed for solid physics research.

In the nuclear industry para-hydrogen is employed to fill bubble chambers.

In electrical power plants hydrogen is used as a coolant gas in turbogenerators.

Hydrogen is used for synthesis of ammonia.

Hydrogen is used as a reagent to produce high-purity water.
Hydrogen is used as fuel in fuel cell applications.

Hydrogen is used as component in gas mixtures.

Physical data

Molecular weight		2.01γ		
Boiling point	at 1.013 bar [°C]	−2β2.7γ	at 1αβ psi [°F] at 1 atm, 70 °F [lb/ft³] at 32 °F [psi] at 70 °F [psi]	−α22.9β
Density	at 1.013 bar, 1β °C [kg/m³]	0.08β2		0.00β
Vapour pressure	at 0 °C [bar]	–		–
	at 20 °C [bar]	–		–
Flammability range in air [% volume]		α.0 – 77.0		

Nitrogen N2

CAS: 7 7 27-37- 9
EC: 231-783-9
UN: 10yy (Compressed) 1977
(Refrigerated liquid)
UN: R-728

Purity grade	Purity	Impurities [ppm]												
		H0□	O□	CnHm	CO	CO□	H□	Ar	NOx	SO□		Odour		Halocarbons
HiQ® Nitrogen 4.6	≥99.99y %	≤β		-	-	-		-	-	-				-
HiQ® Nitrogen 5.0	≥99.999 %	≤3			-	-		-	-	-	NO			-
HiQ® Nitrogen 5.0 Zero	≥99.999	≤3	≤1		≤1	≤β≤1		-	-	≤0.1	≤0.1	-		-
HiQ® Nitrogen 5.5 ECD	% ≥99.999β	≤1	≤0.β	≤0.1	≤0.β	≤2≤0.β	-					-	-	≤1
						≤2≤0.2	≤1	≤0.1	≤0.β	≤0.β	≤1	-	-	ppb
HiQ® Nitrogen 5.5 CEM Zero	% ≥99.999β						-	-				-	-	≤0.1	-	-	-
HiQ® Nitrogen 6.0	%		≤0.β	≤0.β	≤0.1	≤0.1	≤0.1	≤0.β		-		-	≤0.02	-	-	-
HiQ® Nitrogen 7.0	≥99.9999 %	≤β0	≤30	≤30	≤30	≤30			-	-	-			-	N/D	≤1
HiQ® Nitrogen Euro 6 Raw	≥99.99999	ppb	ppb	ppb	ppb	ppb	ppb	-	-	-					N/D	ppb
HiQ® Nitrogen Euro 6 Dilute	% ≥99.999	≤3		≤2≤0.β		≤1	-	-≤0.β %								-
	% ≥99.9999				≤1	≤0.β	≤0.β	≤0.0β	≤0.1	≤0.1	≤β≤300		≤β≤300			-
VERISEQ® Process Nitrogen (pharmaceutical grade)	% ≥99.β % ≥99.999 %	≤β	≤3		≤β	≤3				-						
VERISEQ®Research Nitrogen (pharmaceutical grade)						≤1										

Legend: N/D = Not Detectable

Typical filling pressure: 1β °C: 200 bar(g)/ 70 °F: 2,800 psi(g)

Typical packages

Cylinders	Bundles	Drum tanks	ISO tanks	Tube trailer	Road tanker
•	•				Cryogenic liquid

Typical ancillary equipment

Pressure control valves	Gas distribution panels/manifolds	Liquid flow control valves	Customised distribution systems
•	•	Cryogenic liquid	Consult local team

Characteristics
Colourless and odourless gas. Asphyxiant in high concentrations. Gas density is slightly lighter than air.

Hazard classifications
Globally Harmonized System of classification of chemicals (GHS)
Proposed by the Industry
Signal word: WARNING



H-statements:
Compressed Gas → H280 – Contains gas under pressure; may explode if heated; Refrigerated Gas → H281 – Contains refrigerated gas; may cause cryogenic burns or injury; EIGA-As – Asphyxiant in high concentrations.

Transport of dangerous goods



ADR Class 2, 1A (Compressed)
3A (Refrigerated liquid)



DOT Class 2.2

CHEM CONSTR ENERGY FOOD MANUF MEDICAL METAL OEM PETRO PHARMA R&D SEMI

Source
Nitrogen is produced in large quantities at air separation plants which liquefy and subsequently distil air into nitrogen, oxygen and argon. If very high purity nitrogen is required the nitrogen produced may need to go through a secondary purification process. The lower range of nitrogen purities can also be produced with membrane techniques, and medium to high purities with pressure swing adsorption (PSA) techniques.

Applications
Nitrogen is used in large quantities in the chemical industry for blanketing, purging and pressure transfer of flammable chemicals.

High-purity nitrogen is used in large quantities by the semiconductor industry as a purge or carrier gas as well as for blanketing equipment such as furnaces when not in production.

Nitrogen is used as a purge gas.

Nitrogen is commonly used as carrier gas in gas chromatography.

Nitrogen is used as zero gas for analytical instruments.
Nitrogen is commonly used as a balance gas in mixtures.
Nitrogen is used in the electronic industry for inerting of epitaxial reactors.

Nitrogen is used in mixtures with carbon dioxide for modified atmosphere packaging (MAP) of food stuffs.
Nitrogen is used extensively, either pure or, more commonly, in a mixture with a reducing gas such as hydrogen or natural gas, to provide an oxygen free atmosphere during heat treatment of various metals.

Nitrogen is used in the Haber-Bosch process for production of ammonia.

Nitrogen is used as a fire extinguishing gas in mines.

Nitrogen is used to fill tires to reduce wear and limit the risks of blow-outs.
Liquid nitrogen is used in cold traps to improve the efficiency of vacuum pumps by condensing or solidifying residual gases in the vacuum.

Liquid nitrogen may be used for shrink fitting of close tolerance components.

Liquid nitrogen is used to freeze a wide variety of delicate food, such as hamburgers, strawberries, shrimps etc.

Liquid nitrogen may also be used for cryogenic grinding of plastics, rubbers and some other chemicals products.

Liquid nitrogen is used in the nuclear industry, for scientific research.

Liquid nitrogen is used to store biological materials like tissue, cells etc.

Liquid nitrogen is also used for cryo surgery.

Liquid nitrogen is used in the area of superconductivity.

Nitrogen is used in Liquid chromatography-mass spectrometry.

Physical data

Molecular weight		28.01α		
Boiling point	at 1.013 bar [°C]	−19β.8	at 1α.β psi [°F] at 1	−320.α2
Density	at 1.013 bar, 1β °C [kg/m³]	1.18β	atm., 70 °F [lb/ft³] at	0.072
Vapour pressure	at 0 °C [bar]	–	32 °F [psi] at 70 °F	–
	at 20 °C [bar]	–	[psi]	–
Flammability range in air [% volume]		Non combustible		



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Supply Requirements



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